

Physical AI Understanding & Building the Physical World

The **1st Workshop on Physical AI** brings together researchers across vision, graphics, robotics, and generative modeling to advance physics-grounded understanding of the real world — from physical property estimation and 3D/4D reconstruction to differentiable simulation and physically plausible generation. **14 invited talks · panel · 4D reconstruction challenge.**

VENUE

ECCV 2026 · Malmö, Sweden

DATE

September 9, 2026

FORMAT

14 talks · panel · competition

TOPICS OF INTEREST

- Physical scene understanding
- Physics-aware 3D/4D reconstruction
- Generative world models & world simulators
- Spatial & physical reasoning
- Embodied AI & robot learning
- Benchmarks & datasets

COMPETITION

PhysAI Dynamic 4D Reconstruction Challenge

A new synthetic 4D benchmark. **Awards for top teams** + joint technical report.

TRACK 1 4D Reconstruction & Tracking**TRACK 2** Dynamic Video Novel View Synthesis**INVITED SPEAKERS (TENTATIVE · ALPHABETICAL)**

Andreas Geiger
Tübingen



David Novotny
Spatial AI



Haiwen Feng
UC Berkeley



Jiajun Wu
Stanford



Kristen Grauman
UT Austin



Marc Pollefeys
ETH Zurich



Matthias Nießner
TU Munich



Ming-Yu Liu
NVIDIA



Vincent Sitzmann
MIT

ORGANIZED BY

Oxford VGG · NTU PVG · NTU MMLab · ETH Zurich · Google DeepMind · NAVER LABS Europe

Join the challenge & the conversation.

Contact: yushi@robots.ox.ac.uk

Full details, schedule & challenge → [workshop website](#)

PhysAI 2026

ECCV 2026 · Malmö
1st Workshop on Physical AI